

GPU Batch Lunar Gravity Validation

Fixed-step GPU propagation vs an adaptive reference

2026-05-28 23:42 UTC

Parameter	Value
Scenarios	100
Sampling method	random
Scenario seed	42
Scenario mode	near_circular_altitude
Inclination sampling	uniform_deg
Altitude range	100 - 1000 km
Duration	1 days
GPU integrator	medium (fixed step 60 s)
Truth workers	4
Output cadence	60 s
Precision	float32
Truth	SH200 DOP853
GPU models	sh20_dt60,sh30_dt60,sh100_dt60,sh200_dt60,st_lrps_dt30,st_lrps_dt60
Frame mode	match_dynamics_engine

Reference note: the truth trajectories are a high-accuracy adaptive numerical reference, not physical ground truth. GPU fixed-step models carry both model and integration error.

Executive Summary

Metric	Value
Most accurate GPU model	GPU_SH100_DT60_RK4
Fastest GPU model	GPU_SH20_DT60_RK4
ST-LRPS closest by median RMS	n/a
ST-LRPS median-equivalent status	n/a
ST-LRPS closest by P95 RMS	n/a
Best comparison set scenario	346
Representative comparison set scenario	18
Worst comparison set scenario	67

Accuracy Ranking

Sorted best-to-worst by median RMS position error.

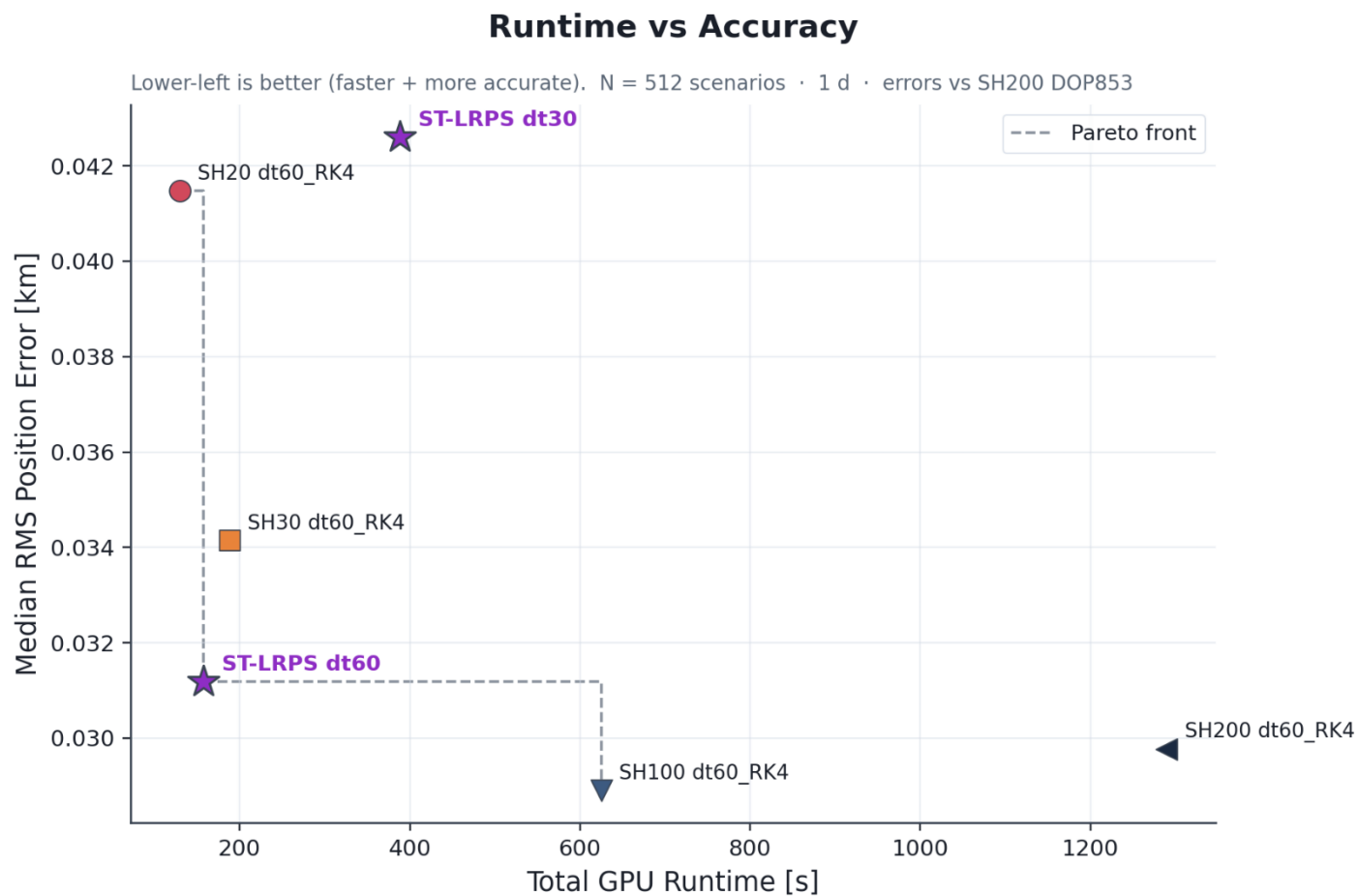
Model	Median RMS [km]	P95 RMS [km]	Max RMS [km]	Median Along [km]
GPU_SH100_DT60_RK4	0.0289	0.0836	0.1335	0.0289
GPU_SH200_DT60_RK4	0.0298	0.0850	0.1397	0.0297
ST_LRPS_DT60	0.0312	0.0799	0.1164	0.0310
GPU_SH30_DT60_RK4	0.0341	0.1104	0.3998	0.0340
GPU_SH20_DT60_RK4	0.0415	0.2698	0.9723	0.0394
ST_LRPS_DT30	0.0426	0.1184	0.1835	0.0424

Runtime

Wall-clock runtime and throughput for the GPU fixed-step propagation.

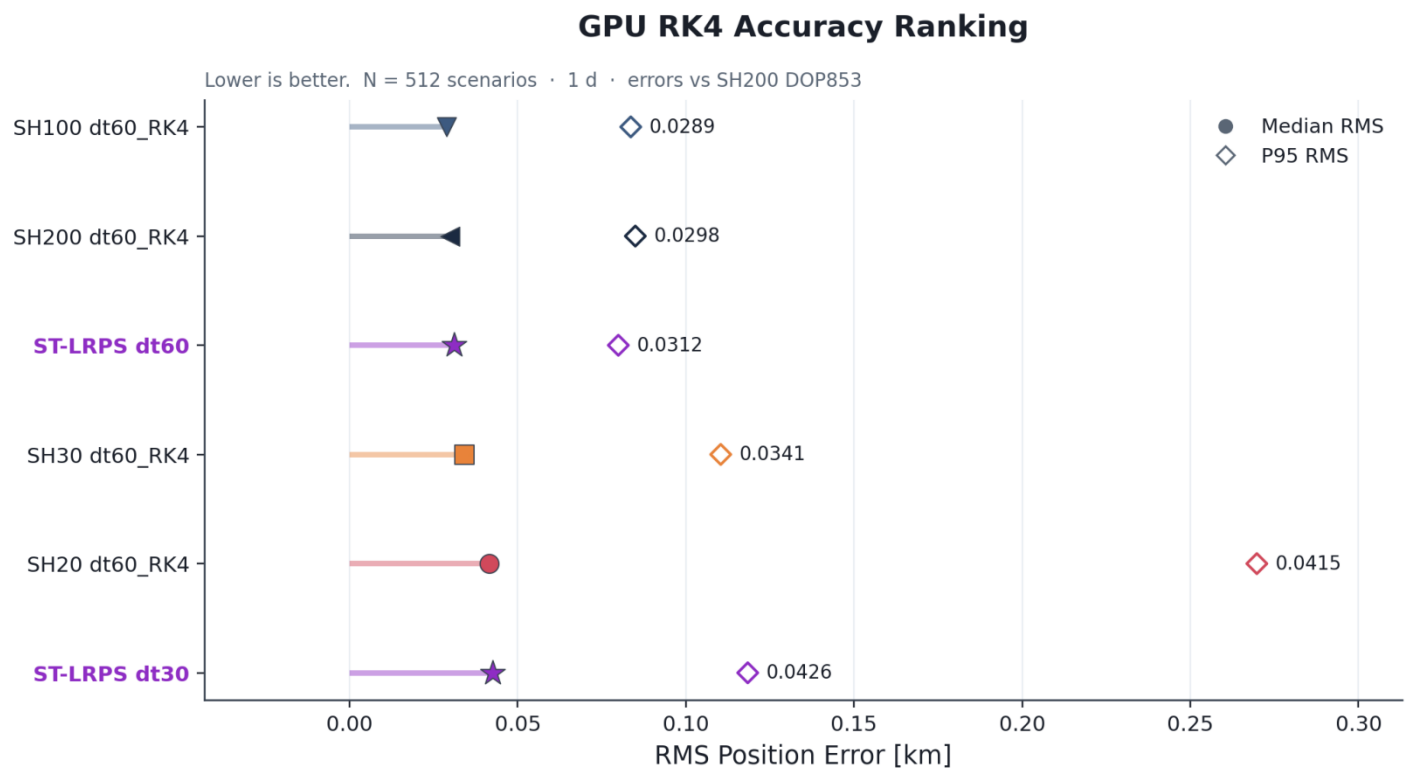
Model	Total [s]	Per scenario [s]	Steps/s	Speedup vs truth
GPU_SH20_DT60_RK4	129.622	0.25317	2844.0	113.29
ST_LRPS_DT60	157.570	0.30775	2339.5	93.20
GPU_SH30_DT60_RK4	188.505	0.36817	1955.6	77.90
ST_LRPS_DT30	388.377	0.75855	949.2	37.81
GPU_SH100_DT60_RK4	625.196	1.22109	589.6	23.49
GPU_SH200_DT60_RK4	1289.620	2.51879	285.9	11.39

Figure



Runtime-accuracy tradeoff across GPU models. Lower-left is better (faster and more accurate); the dashed line marks the Pareto front. N = 100 scenarios over 1 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

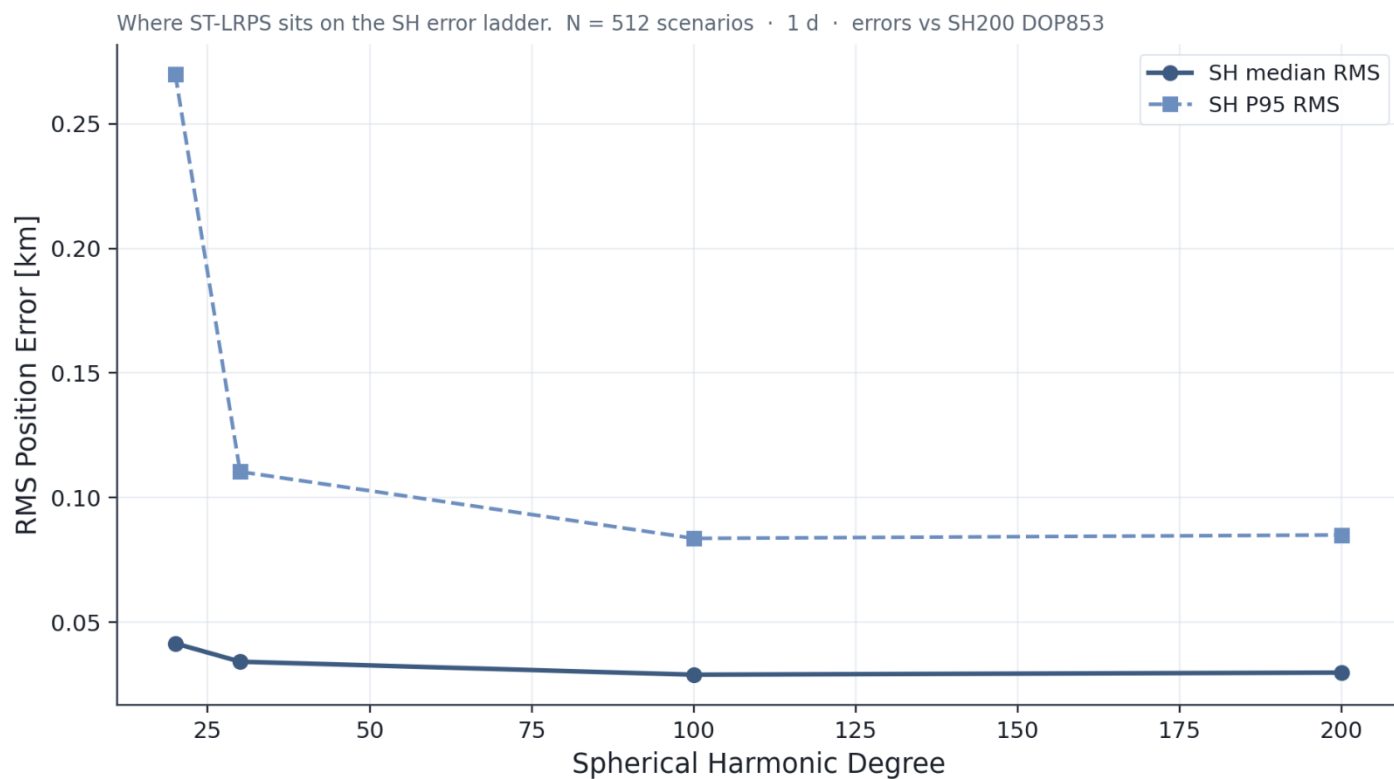
Figure



Per-model accuracy ranking (lollipops = median RMS, open diamonds = P95 RMS), sorted best-to-worst; ST-LRPS is highlighted. Lower is better. N = 100 scenarios over 1 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

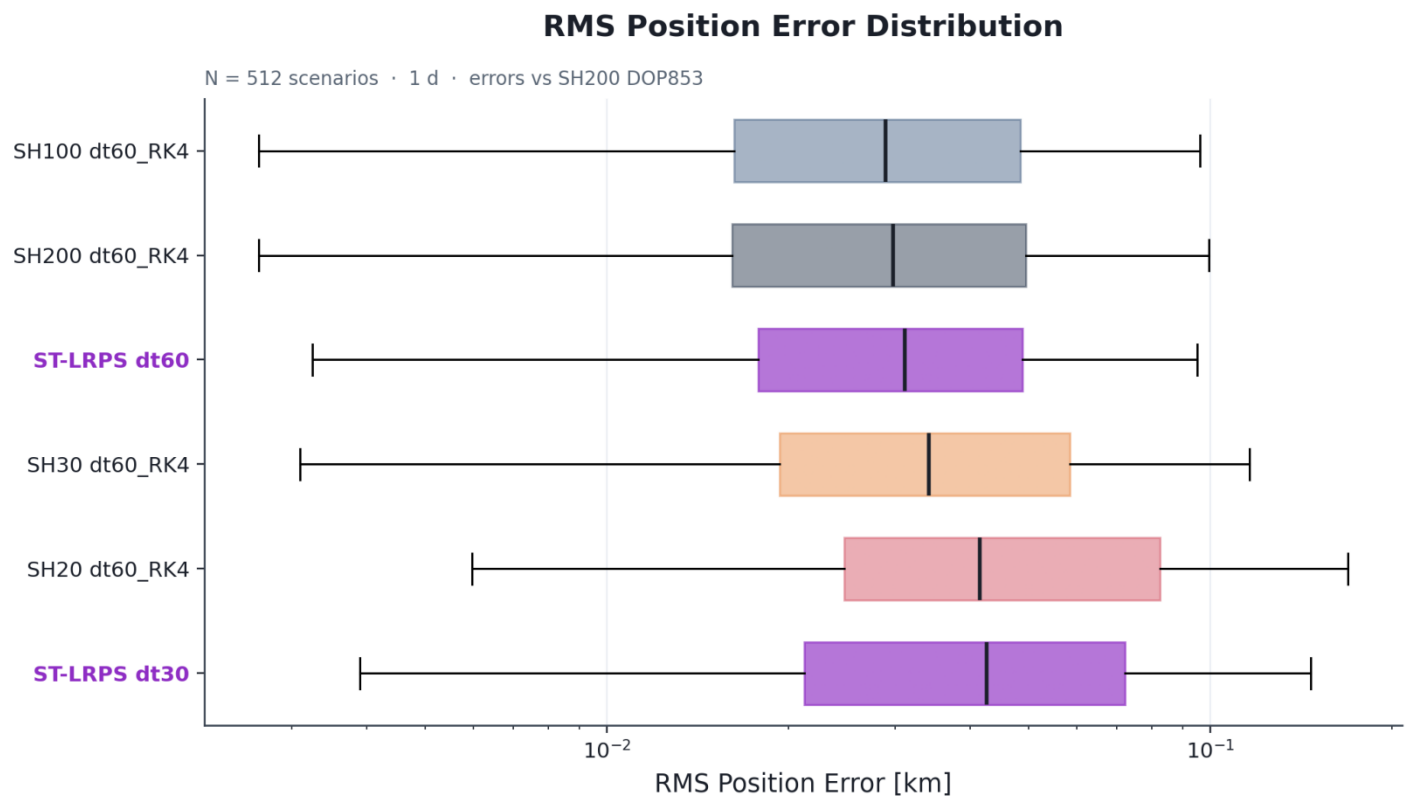
Figure

ST-LRPS Equivalent Spherical-Harmonic Degree



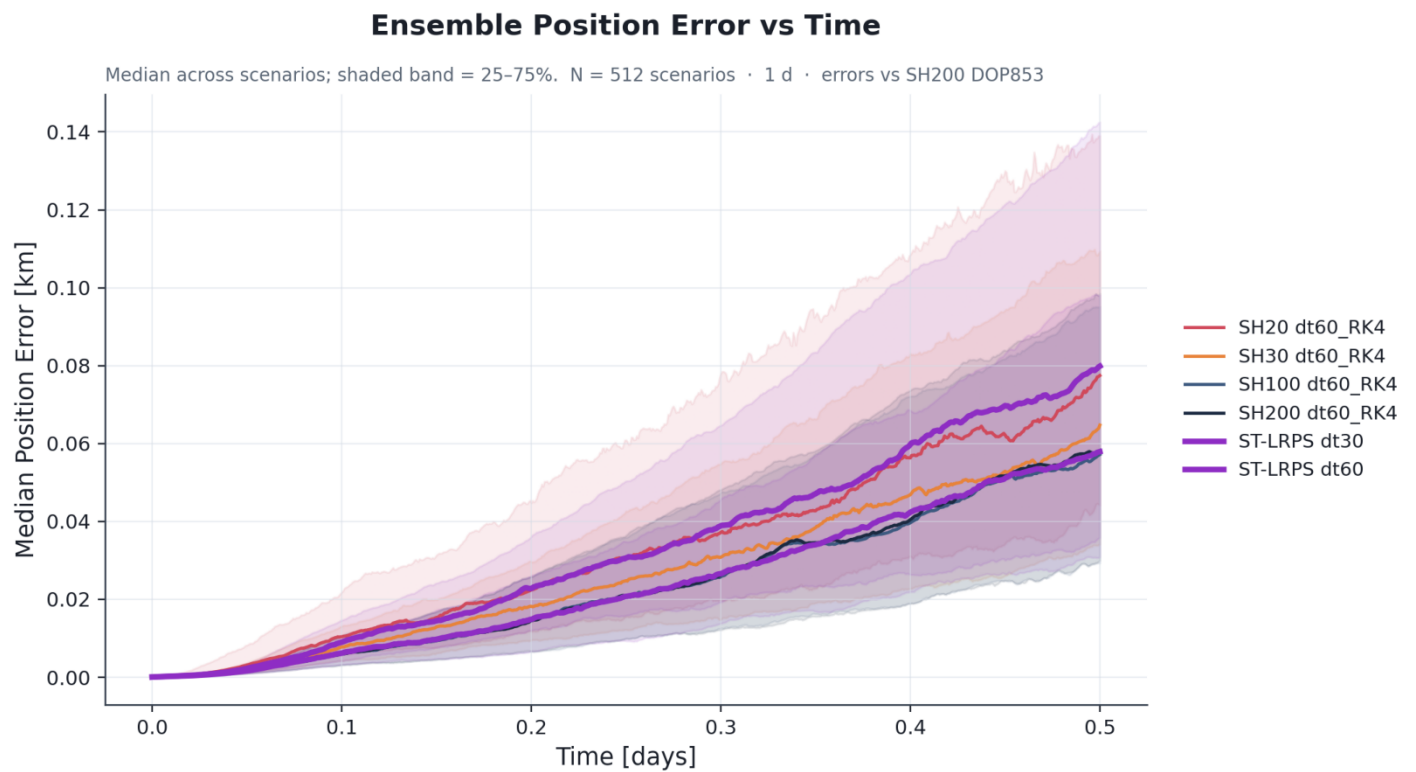
ST-LRPS equivalent SH-degree estimate by interpolating median RMS error across the spherical-harmonic baselines. N = 100 scenarios over 1 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



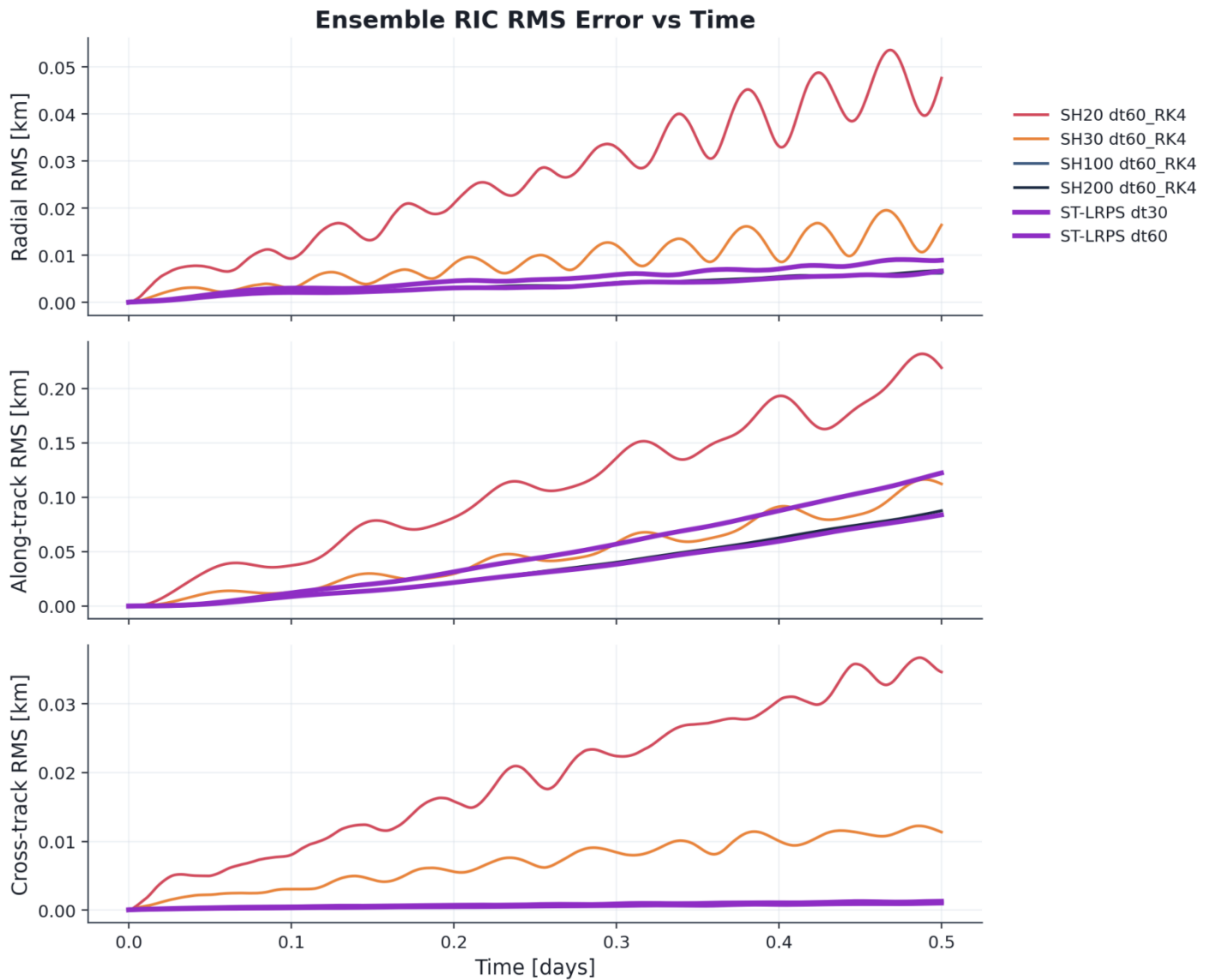
Distribution of per-scenario RMS position error per model. N = 100 scenarios over 1 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



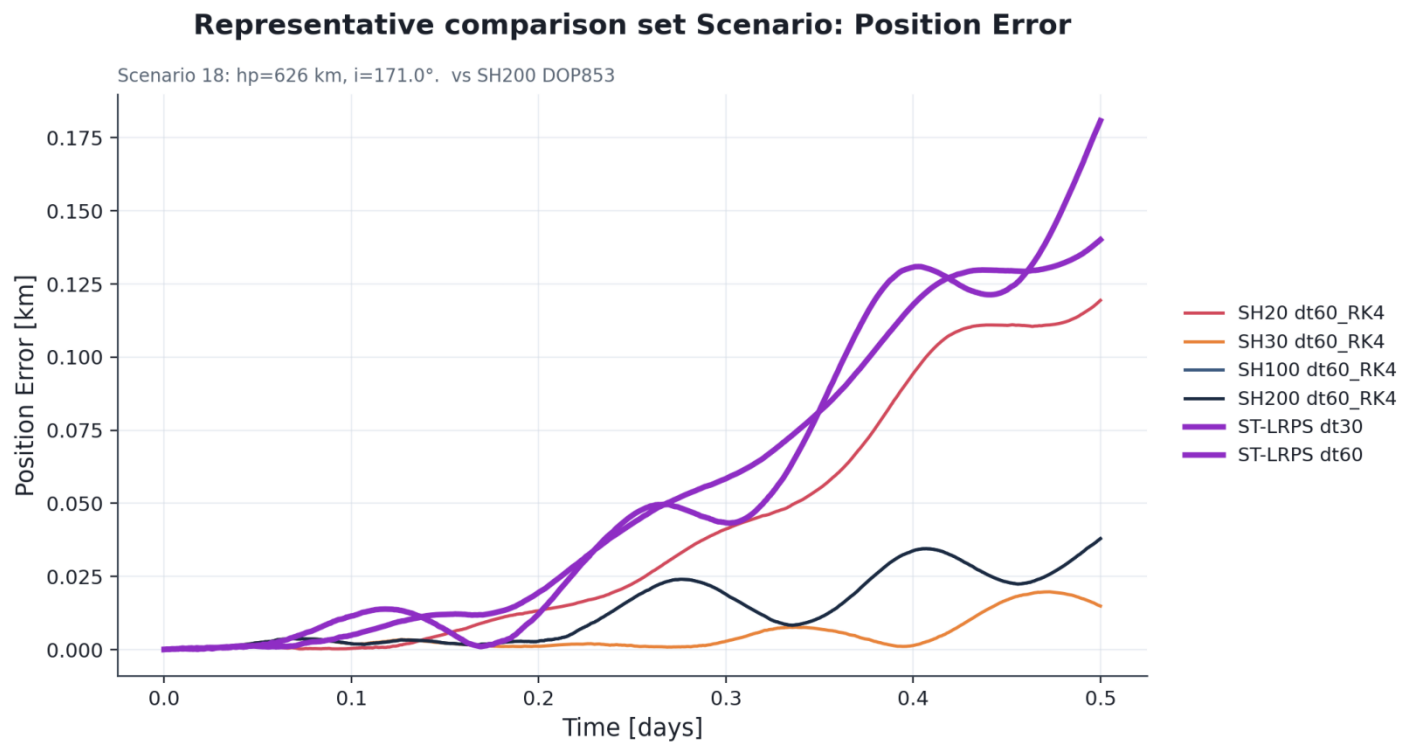
Ensemble median position error vs time (shaded band = 25-75% across scenarios). N = 100 scenarios over 1 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



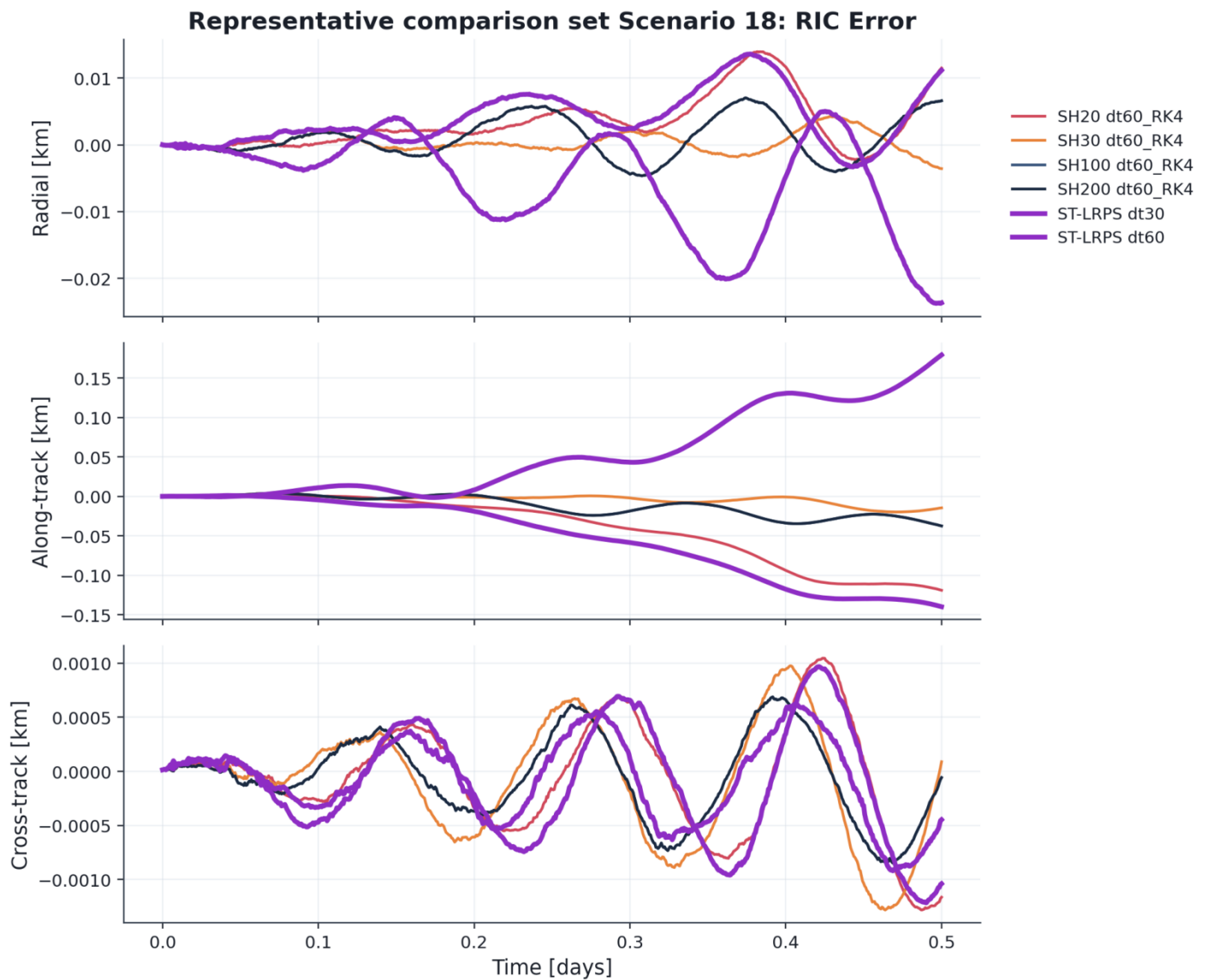
Ensemble radial/along-track/cross-track RMS error vs time. $N = 100$ scenarios over 1 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



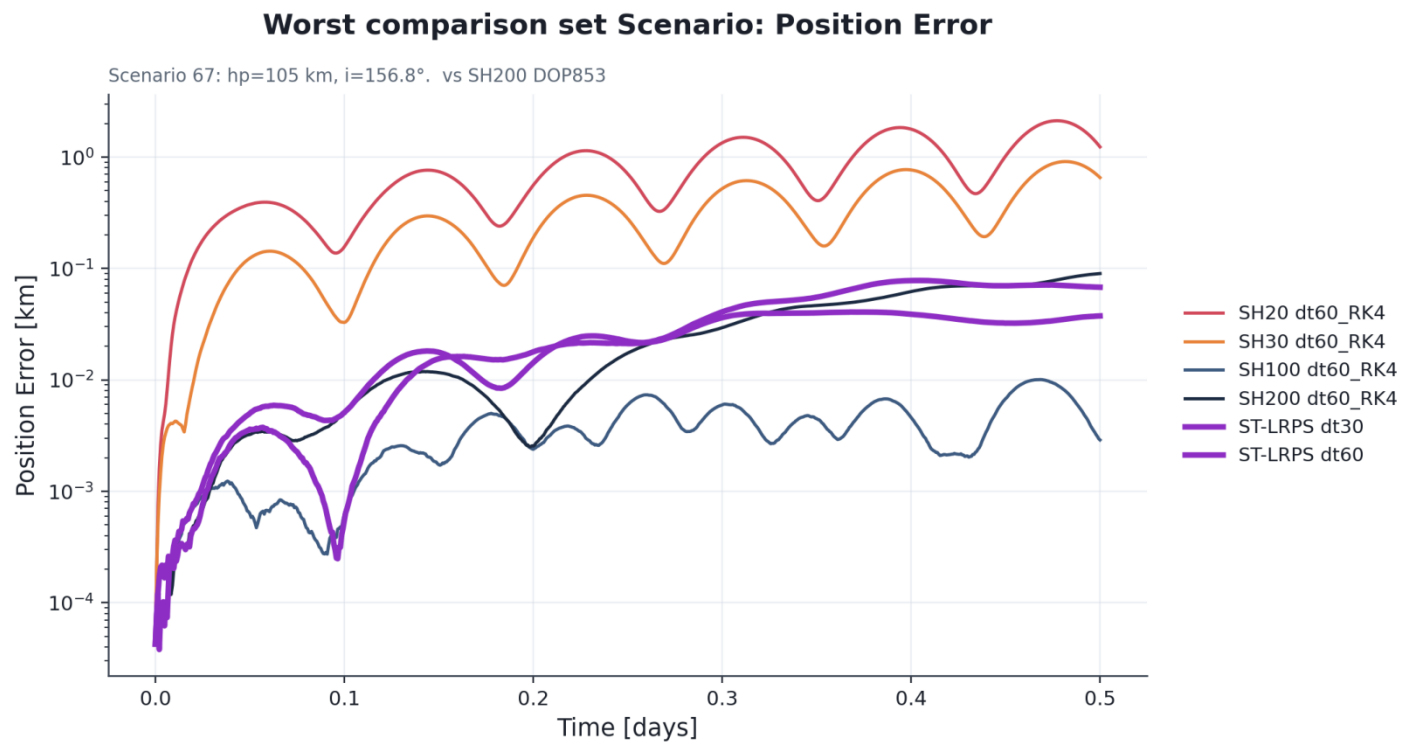
Representative scenario: position error vs time. N = 100 scenarios over 1 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



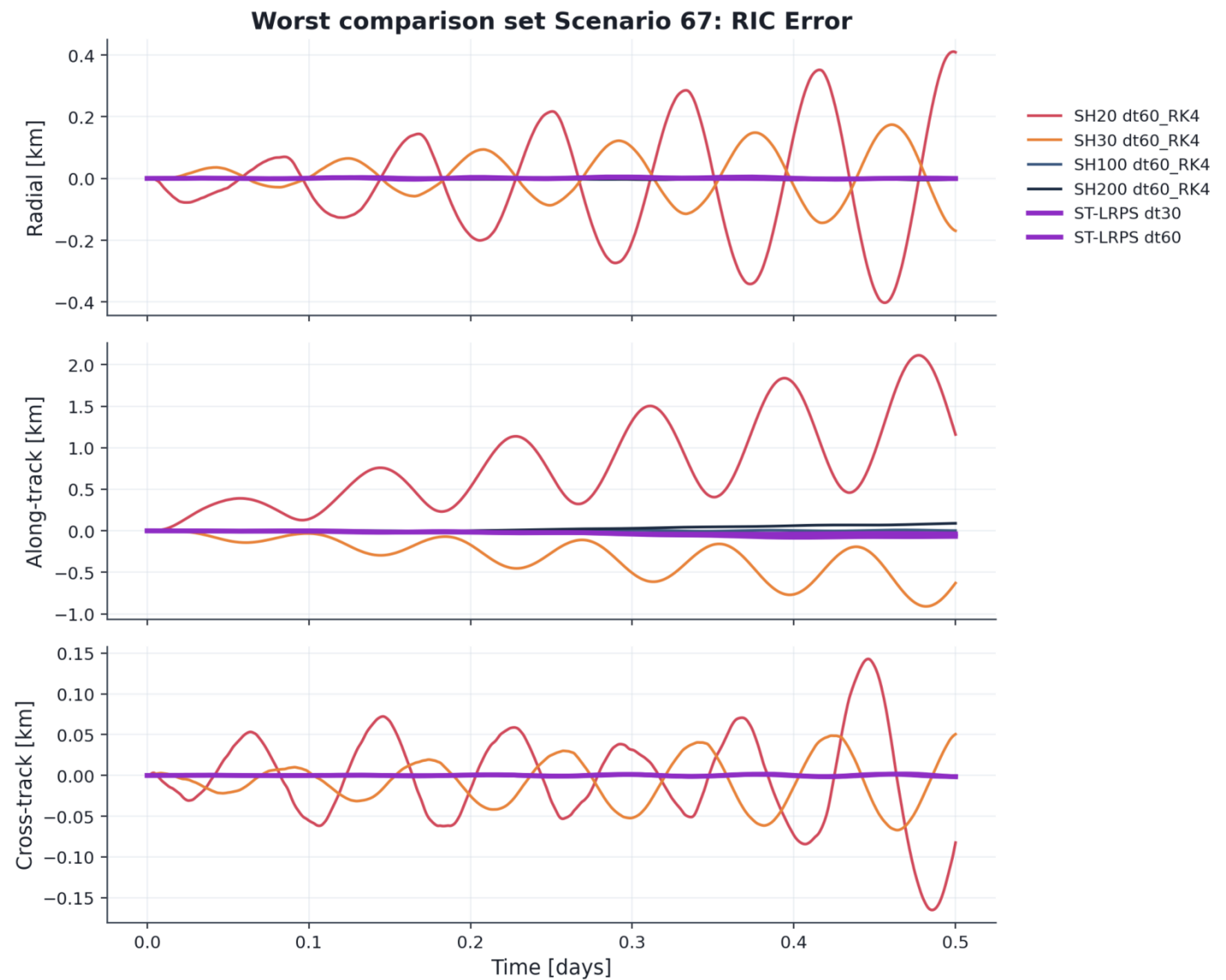
Representative scenario: radial/along-track/cross-track error vs time. N = 100 scenarios over 1 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



Worst-case scenario: position error vs time. $N = 100$ scenarios over 1 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



Worst-case scenario: radial/along-track/cross-track error vs time. N = 100 scenarios over 1 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Notes & Caveats

- The truth trajectories are a high-accuracy adaptive numerical reference, not physical ground truth.
- GPU fixed-step SH models include both spherical-harmonic truncation error and integration error.
- ST-LRPS includes surrogate-model error plus integration error.
- Frame mode: `match_dynamics_engine`.